

PHASE III – LOGICAL DATABASE DESIGN & NORMALIZATION

Project: Umurenge SACCO Savings & Loan Management System

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1. Unnormalized Form (UNF)

Initially, all data was stored in one large table with repeating fields (multiple loans, deposits, and repayments for one member). This caused duplication, update anomalies, and made it impossible to enforce rules such as blocking overdue members or auditing individual transactions.

2. First Normal Form (1NF)

Repeating groups were removed by splitting the data into six separate entities: MEMBERS, LOAN_TYPES, LOANS, SAVINGS_DEPOSITS, LOAN_REPAYMENTS, and AUDIT_LOG. Each cell in every table now holds a single atomic value, and each table has a defined primary key.

3. Second Normal Form (2NF)

Every table uses a single-column surrogate primary key (member_id, loan_id, loan_type_id, deposit_id, repayment_id, log_id). Because no table has a composite primary key, there are no partial dependencies — every non-key column depends on the entire primary key. All six tables therefore satisfy 2NF.

4. Third Normal Form (3NF)

The tables were checked for transitive dependencies (a non-key column depending on another non-key column rather than directly on the primary key):

- In **LOANS**: **interest_amount** and **total_due** depend on **loan_id**, not on **principal_amount** or **duration_months** alone.
- In **MEMBERS**: **village**, **cell**, and **sector** depend directly on **member_id**, not on each other.
- In **LOAN_TYPES**: **interest_rate** and **max_amount** depend only on **loan_type_id**.

No table has a non-key column that determines another non-key column, so no transitive dependencies exist.

Conclusion: All six tables are fully compliant with Third Normal Form (3NF).

5. Entity Summary

Entity	Primary Key	Purpose
MEMBERS	member_id	Stores member identity and registration details.
LOAN_TYPES	loan_type_id	Defines the loan products offered and their terms.
LOANS	loan_id	Tracks each loan issued to a member, its interest, and status.
SAVINGS_DEPOSITS	deposit_id	Records each savings deposit made by a member.
LOAN_REPAYMENTS	repayment_id	Records each repayment made against a loan.
AUDIT_LOG	log_id	Logs changes made across tables for transparency and accountability.